TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Subhajeet Rath	Email	subhajeet.rath@tno.nl
Institute Name	Nederlandse Organisatie voor Toegepast Natuurwetenschappelijk Onderzoek (TNO), Powertrains Department			
Address	Automotive Campus 30, 5708 JZ Helmond, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Predictive moisture-stability assessment and accelerated testing framework for sulfide-based solid electrolytes (SSE)			
Korea's 12 National Strategic Technology Fields	Semiconductors & Displays ( ), Secondary Batteries (√), Advanced Mobility ( ), Next-generation Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cyber Security ( ), Artificial Intelligence ( ), Next-generation Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Battery Materials	Batteries	Secondary Batteries
Concept of Proposed Technology	The proposed technology aims to develop a predictive moisture-stability assessment and accelerated testing framework for sulfide-based solid electrolytes (SSEs) used in all-solid-state batteries. While sulfide SSEs offer high ionic conductivity and favorable processing characteristics, their sensitivity to moisture remains a major barrier to safe handling, scalable manufacturing, and industrial deployment. Exposure to humidity can induce hydrolysis reactions, H_2S formation, and interfacial degradation, leading to increased impedance and degraded			

	electrochemical performance [12]. A quantitative and reproducible method is therefore required to translate dry-room specifications into electrochemical performance and safety criteria. The core concept is to establish a lab-to-field qualification framework that systematically links controlled moisture exposure conditions—defined by humidity or dew point, temperature, and exposure time—to measurable electrochemical degradation metrics. Using climate-controlled test environments, representative electrolyte samples will be subjected to reproducible exposure matrices reflecting realistic handling and manufacturing scenarios. Electrochemical Impedance Spectroscopy (EIS) and complementary electrochemical measurements will be applied to track changes in ionic transport and interfacial resistance as a function of moisture dose [1][2]. The resulting datasets will be analyzed using physics-informed and data-driven models to identify sensitive degradation indicators and predict performance drift under given exposure conditions. These models enable the derivation of moisture-stability thresholds and acceptance criteria, translating laboratory measurements into actionable guidance such as allowable dew-point windows and exposure limits during processing and handling [4][12]. The outcome of the proposed technology is a standardized, reproducible qualification protocol, including defined test conditions, electrochemical KPIs, and predictive decision support. This transforms moisture sensitivity from an empirical risk into a quantified and manageable design parameter, reducing scale-up uncertainty and improving industrial readiness of sulfide-based solid electrolytes [2][4]
Rationale for Joint R&D Project	Sulfide-based solid electrolytes are attractive for all-solid-state batteries but their moisture sensitivity creates manufacturing and handling bottlenecks, including potential toxic gas formation and impedance growth. [12] TNO contributes a combination of (i) climate-controlled testing infrastructure with EIS and hardware-in-the-loop capable equipment, and (ii) model-based interpretation workflows that link measured impedance and performance changes to predictive health and lifetime models. [1][2][4] This complementarity enables a joint approach: the Korean SME partner provides electrolyte material know-how and processing constraints, while TNO provides controlled exposure and electrochemical test facilities plus modelling and algorithm expertise to turn measurements into a repeatable qualification protocol and predictive guidance for scale-up. [2][13]
Key Words	1. Sulfide solid electrolyte 2. Moisture stability 3. Electrochemical Impedance Spectroscopy (EIS) 4. Interfacial impedance; qualification protocol 5. Predictive modelling

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	Establish a quantitative and reproducible moisture-stability qualification protocol for the Korean SME partner sulfide-based solid electrolytes, including: • A controlled exposure matrix (humidity or dew point, temperature, time) aligned with dry-room and manufacturing conditions. [12][1] • Electrochemical KPIs and measurement procedures (EIS spectra, DC resistance, and derived conductivity or interfacial resistance indicators) with repeatability assessment. [12][1][2] • A predictive model that maps exposure conditions to electrochemical KPI drift, supporting process windows and handling guidelines for scale-up. [4]			
R&D Target Specifications	Target specifications are to be defined jointly after baseline material screening, but the protocol will at minimum provide: • Defined test coupons or cell formats for electrolyte measurement (for example, pressed pellets or representative interfaces). • A standardised EIS and cycling test sequence with clearly defined boundary conditions (temperature, pressure if applicable, and measurement frequency). [12] • Acceptance criteria expressed as KPI limits and confidence intervals derived from repeatability and sensitivity analyses.			
R&D Plan	Proposed 24-month work plan (timeline and milestones to be aligned with the selected TIPA track): **WP 1 - Baseline and protocol design (Month 0–8)** • Agree sample formats, handling constraints, and baseline electrochemical characterisation under dry conditions. • Design the moisture exposure matrix using a controlled, climate-controlled environment and a safety-oriented handling plan. [1][2][12] **WP 2 - Controlled exposure and electrochemical mapping (Month 6–18)** • Execute exposure and recovery cycles, collecting EIS and performance data to quantify impedance growth and transport changes as a function of moisture dose. [12][1][2] **WP 3 - Modelling and lab-to-field translation (Month 16–24)** • Fit physics-informed and data-driven models and translate results into actionable process windows and qualification checks. • Document an SOP and data package to support internal QC and future scale-up discussions. [4]			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) expected roles: • Provide controlled-environment electrochemical testing (including climate-controlled humidity and temperature conditions) and EIS-based diagnostics to systematically assess moisture-induced degradation effects in sulfide-based solid electrolytes. • Design and execute reproducible exposure and measurement protocols, aligned with the Korean SME partner's handling, storage, and manufacturing constraints. • Apply model-based and data-driven interpretation workflows to link measured impedance and performance changes to predictive indicators relevant for qualification and scale-up. • Develop and document predictive models and acceptance criteria that translate laboratory measurements into actionable process windows and moisture-stability guidelines. • Deliver documented datasets, analysis reports, and a draft SOP to support technology transfer, internal quality control, and future industrial deployment by the Korean SME partner.
Expected R&R of Korean SME Partner	Provide sulfide electrolyte samples (reference and development variants) and relevant manufacturing context. Expected tasks: • Define use-case requirements and constraints for handling, storage, and product qualification. • Supply representative samples and batch metadata needed for analysis. • Support interpretation of results relative to formulation and process variables. • Pilot the SOP internally and provide feedback for protocol finalization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected deliverables: • Standardized accelerated moisture-stability test protocol for sulfide solid electrolytes, including KPI definitions and reporting format. • Predictive model or decision-support tool linking exposure conditions to KPI drift and qualification outcomes. • Practical storage and handling guideline structure for production, transport, and partner evaluation. • Impact for the Korean SME partner: • Reduced qualification uncertainty when scaling production and engaging automotive or battery-industry customers. • Earlier detection of moisture-related risks and more robust product positioning for solid-state battery markets.

6. References

(1) TNO, "Sustainable batteries with advanced battery management" (webpage). Includes Battery Lab temperature range (-40°C to +150°C), humidity control, 300 kW battery cycler, HiL capability, and EIS mention. Accessed 2025-12-16. URL: https://www.tno.nl/en/sustainable/mobility-logistics/batteries/

(2) TNO, "Battery Lab" facility page (webpage). Describes integrated testing and research, hardware-in-the-loop setups, characterisation and safety portfolio including EIS-supported thermal runaway work. Accessed 2025-12-16. URL: https://www.tno.nl/en/technology-science/facilities/battery-lab/

(3) TNO Fast Track, "Batterij Lab: test batterij technologieën of toepassingen" (webpage, Dutch). Lists example test capabilities including controllable humidity climate environments, cell test channels up to 100 A (parallel 1200 A), EIS up to 128 A and 10 kHz, and pack test up to 750 V and 400 A. Accessed 2025-12-16. URL: https://fasttrack.tno.nl/activiteiten/batterijlab-test-optimale-toepassing/

(4) TNO publication (PDF), "Battery Modelling & Battery Management Systems" (file name TNO-2024-battery.pdf). Describes BMS toolchain elements including SoX estimation, parameter identification, validated models in MATLAB/Simulink, and BMST. Accessed 2025-12-16. URL: https://publications.tno.nl/publication/34697732/VSq0gb/TNO-2024-battery.pdf

(5) A. Dadikozyan et al., "Modular Modelling and Simulation Environment for Battery Design and Analysis - Battery Modular Simulation Tool (BMST)" (conference paper; PDF via ResearchGate). Accessed 2025-12-16. URL: https://www.researchgate.net/publication/380128857_Modular_Modelling_and_Simulation_Environment_for_Battery_Design_and_Analysis_-_Battery_Modular_Simulation_Tool_BMST

(6) NEXTBMS project website (Battery remaining useful life prediction). Accessed 2025-12-16. URL: https://www.nxtbms.eu/

(7) Eindhoven University of Technology (TU/e) researcher profile: Steven Wilkins. Describes background and TNO Powertrains affiliation. Accessed 2025-12-16. URL: https://www.tue.nl/en/research/researchers/steven-wilkins

(8) ORCID record: Steven Wilkins (0000-0001-9498-2321). Includes biography and TNO affiliation. Accessed 2025-12-16. URL: https://orcid.org/0000-0001-9498-2321

(9) TNO Repository entry: Rosca, B. and Wilkins, S., "Enhanced battery model including temperature effects" (conference paper, 2013). Accessed 2025-12-16. URL: https://repository.tno.nl/SingleDoc?docId=39157

(10) Geerts, D. et al., "Charge Scheduling of Electric Vehicle Fleets: Maximizing Battery Remaining Useful Life Using Machine Learning Models" Batteries 2024, 10(2), 60. Published 2024-02-15. DOI: https://doi.org/10.3390/batteries10020060 (open access PDF mirror used: https://pure.tue.nl/ws/portalfiles/portal/334231272/batteries-10-00060-v2.pdf ; accessed 2025-12-16).

(11) TNO Repository entry: "Model identification for thermal modelling of a battery pack" (conference paper, 2017) by Wilkins, S. et al. Accessed 2025-12-16. URL: https://repository.tno.nl/SingleDoc?docId=46763

(12) Yersak, T.A. et al., "Moisture Stability of Sulfide Solid-State Electrolytes" Frontiers in Energy Research, published 2022-05-09. DOI: 10.3389/fenrg.2022.882508. URL: https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2022.882508/full (accessed 2025-12-16).

(13) TNO Newsroom, "TNO opens test lab for next generation battery technology" published 2024-05-13. URL: https://www.tno.nl/en/newsroom/2024/05/tno-opens-test-lab-next-generation/ (accessed 2025-12-16).

(14) Holst Centre, "Batteries" expertise page. Notes upscalable dry coating and thin-thick film deposition for battery components. Accessed 2025-12-16. URL: https://holstcentre.com/expertise/components-devices-systems/batteries/

(15) Holst Centre, "Method for effective production of electrolytes for solid-state batteries" (webpage). Describes LiPON electrolyte via atmospheric-pressure large-area spatial ALD. Accessed 2025-12-16. URL: https://holstcentre.com/insights/news/method-for-effective-production-of-electrolytes-for-solid-state-batteries/

(16) Holst Centre, "Next-gen Equipment for Batteries and Battery-materials" (webpage). Accessed 2025-12-16. URL: https://holstcentre.com/insights/news/next-gen-equipment-for-batteries-and-battery-materials/

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization